

Assignment 4: Model-Based RL

Due December 3rd, 11:59 pm

2 Analysis

Problem 2.1

$$\begin{aligned}
 Q^\pi - \hat{Q}^\pi &= (I - \gamma P^\pi)^{-1} r - (I - \gamma P^\pi)^{-1} (I - \gamma P^\pi) \hat{Q}^\pi \\
 &= (I - \gamma P^\pi)^{-1} (I - \gamma \hat{P}^\pi) \hat{Q}^\pi - (I - \gamma P^\pi)^{-1} (I - \gamma P^\pi) \hat{Q}^\pi \\
 &= (I - \gamma P^\pi)^{-1} [(I - \gamma \hat{P}^\pi) - (I - \gamma P^\pi)] \hat{Q}^\pi \\
 &= (I - \gamma P^\pi)^{-1} (\gamma P^\pi - \gamma \hat{P}^\pi) \hat{Q}^\pi \\
 &= \gamma (I - \gamma P^\pi)^{-1} (P^\pi - \hat{P}^\pi) \hat{Q}^\pi \\
 &= \gamma (I - \gamma P^\pi)^{-1} (P\Pi - \hat{P}\Pi) \hat{Q}^\pi \\
 &= \gamma (I - \gamma P^\pi)^{-1} (P - \hat{P}) \Pi \hat{Q}^\pi \\
 &= \gamma (I - \gamma P^\pi)^{-1} (P - \hat{P}) \hat{V}^\pi
 \end{aligned}$$

Problem 2.2

1. Correct. Although $\hat{V}^\pi$ depends on learned dynamics, the Hoeffding's inequality is only applied to bound $\|P(\cdot|s, a) - \hat{P}(\cdot|s, a)\|_1$ which is agnostic to $\hat{V}^\pi$.
2. Incorrect. For any s, a and N samples $\{s'_i\}$ from $P(\cdot|s, a)$, since $\hat{V}^\pi$ depends on learned dynamics for all states, $X_i = I_{s'_i} \hat{V}$ are not independent. So, Hoeffding's inequality doesn't apply.
3. Correct (subject to a small typo correction). Since V^* doesn't depend on the learned dynamics, the reasoning given in Hint 3 is applicable. Specifically, fix s, a , let s'_i denote the i -th independent sample from $P(\cdot|s, a)$ and let $X_i := I_{s'_i} V^*$. Then, as pointed out in Hint 3, we have $\mathbb{E}[X_i] = P(\cdot|s, a) \cdot V^*$ and $\bar{X}_N := \frac{1}{N} \sum_i X_i = \hat{P}(\cdot|s, a) \cdot V^*$. Since V^* is constant independent of the samples, X_i 's are i.i.d, and $0 \leq X_i \leq \|V^*\|_\infty \leq \frac{1}{1-\gamma}$. So, Hoeffding's inequality is applicable and for $\epsilon = \frac{1}{1-\gamma} \sqrt{\frac{1}{2N} \log \frac{2|S||A|}{\delta}}$, we have

$$P(\bar{X}_N \geq \mathbb{E}[X_i] + \epsilon) \leq \exp\left(-\frac{2N\epsilon^2}{\|V^*\|_\infty^2}\right) \leq \frac{\delta}{2|S||A|}$$

And similar results hold for $P(\bar{X}_N \leq \mathbb{E}[X_i] - \epsilon)$. So, applying union bound, we get

$$P(|\hat{P}(\cdot|s, a) \cdot V^* - P(\cdot|s, a) \cdot V^*| \leq \epsilon) = P(|\bar{X}_N - \mathbb{E}[X_i]| \leq \epsilon) \geq 1 - \frac{\delta}{|S||A|}$$

Applying union bound across all s, a , we get that with probability $1 - \delta$, it holds that

$$\|(\hat{P} - P) \cdot V^*\|_\infty = \max_{s, a} |\hat{P}(\cdot|s, a) \cdot V^* - P(\cdot|s, a) \cdot V^*| \leq \frac{1}{1-\gamma} \sqrt{\frac{1}{2N} \log \frac{2|S||A|}{\delta}}$$

4. Incorrect. Similar to item 2, $\hat{V}^*$ depends on learned dynamics so the assumptions for applying Hoeffding's inequality don't hold.

4 Code

Problem 1

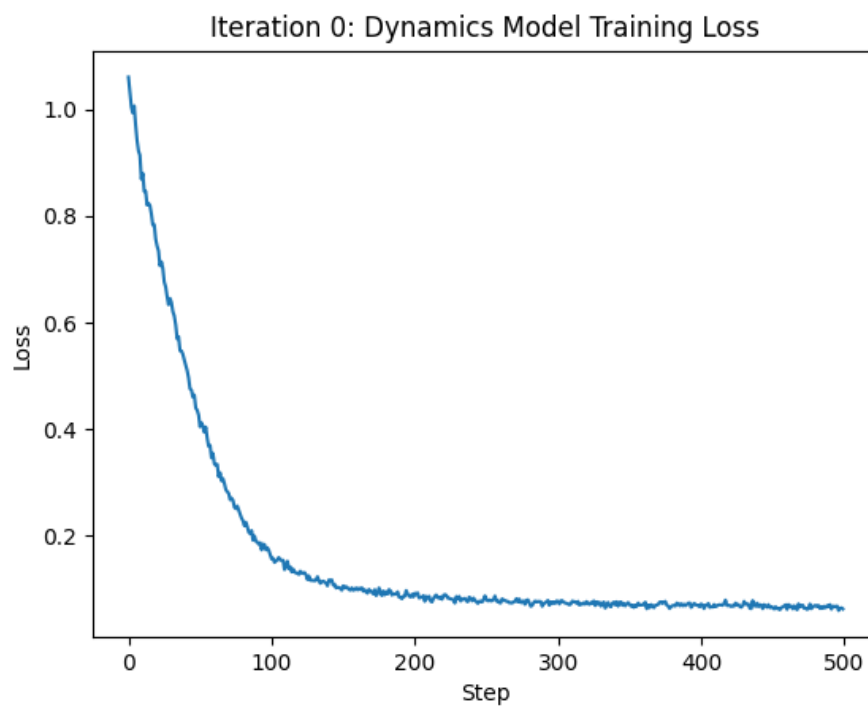


Figure 1: Learning curve for original run

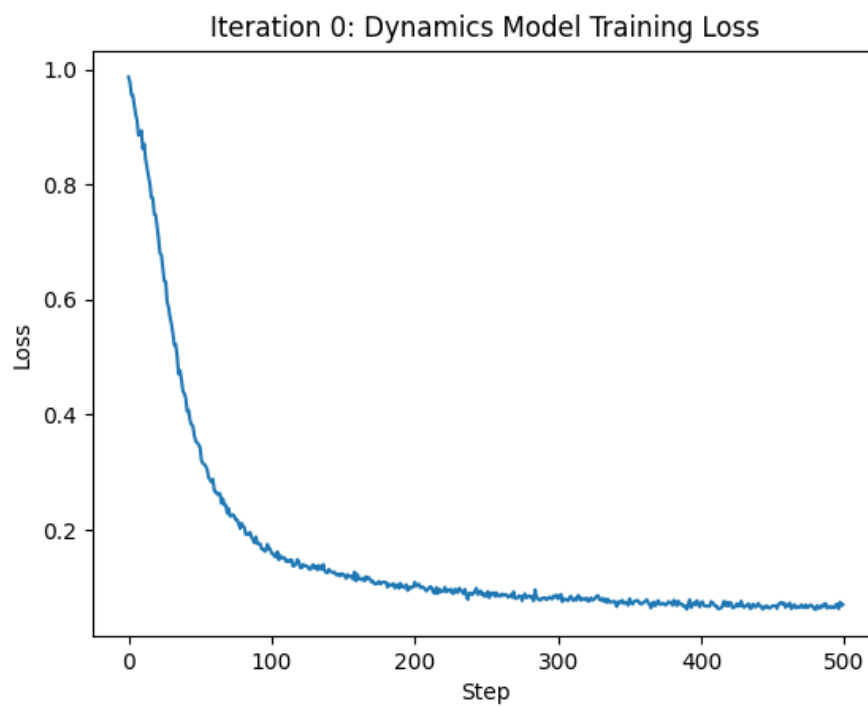


Figure 2: Learning curve for a 3-layer network

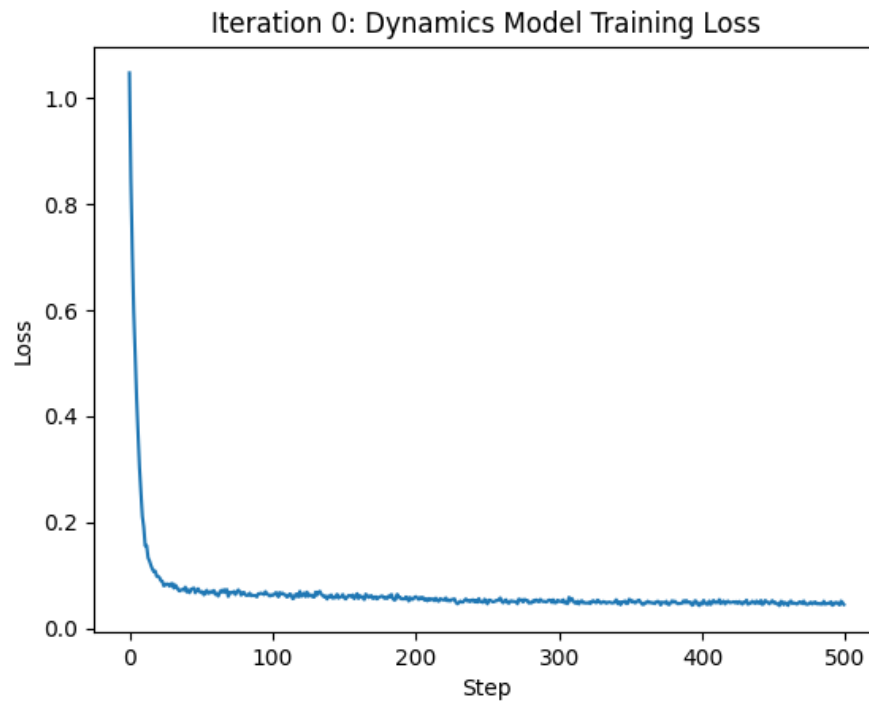


Figure 3: Learning curve for learning rate = 0.01

Problem 2

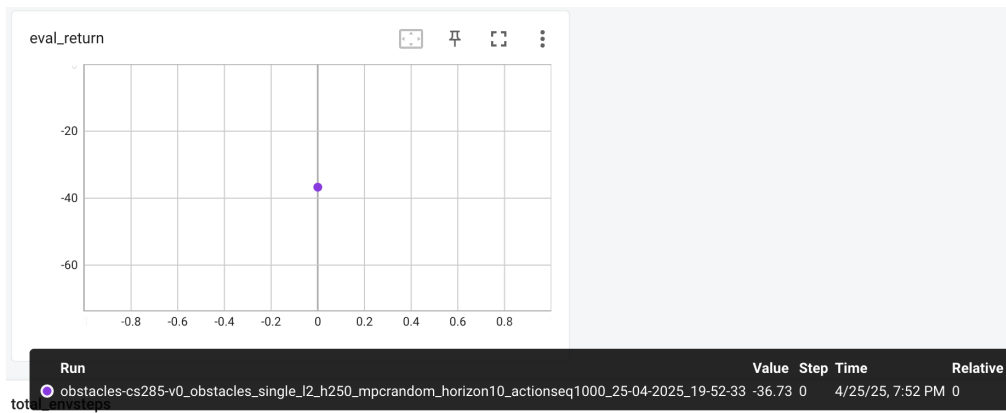


Figure 4: Eval return achieved: -36.73

Problem 3

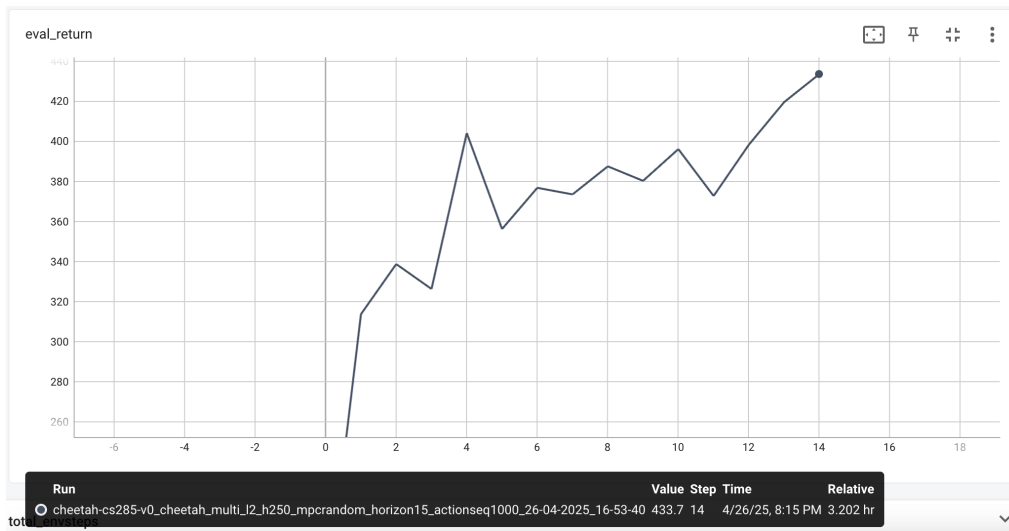


Figure 5: Eval return for Cheetah

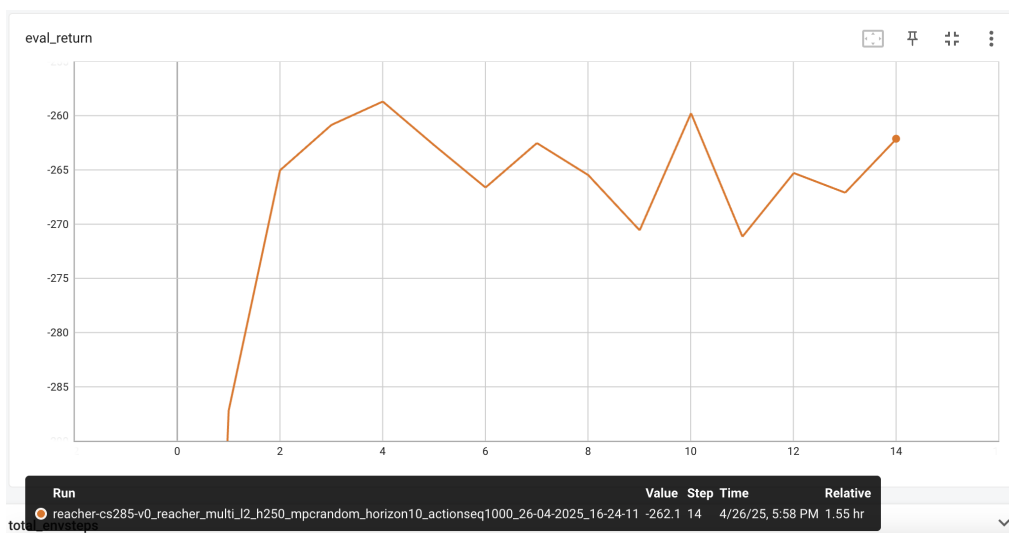


Figure 6: Eval return for Reacher

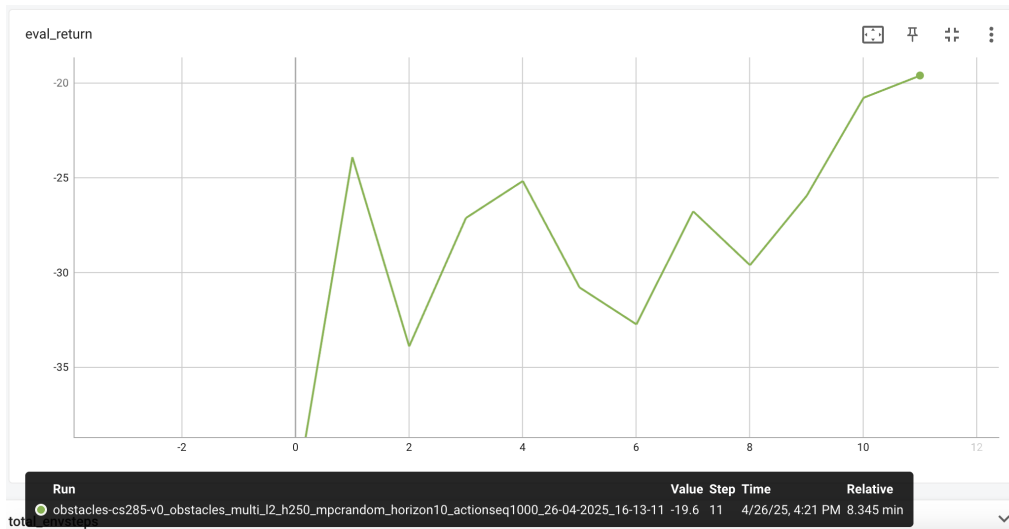


Figure 7: Eval return for Obstacles

Problem 4

Observations with respect to different factors:

- **Horizon:** a shorter horizon of 5 (green line) performed the best, whereas a longer horizon of 20 (orange line) performed the worst. This is because the longer the horizon is, the more likely the model predicted dynamics are inaccurate, which leads to worse estimation of the total reward.
- **Number of action sequences:** doubling the number of action sequences to 2000 achieved the second best performance, whereas reducing it to 500 achieved the second worst. This makes sense because the best among a bigger candidate set is likely better than the best among a smaller set.
- **Ensemble size:** a smaller ensemble size of 1 (pink line) clearly performed worse than the baseline (purple line), whereas a bigger ensemble size of 6 (yellow line) didn't do better than baseline. This seems to suggest that there's a diminishing return point of increasing the size of the ensemble.

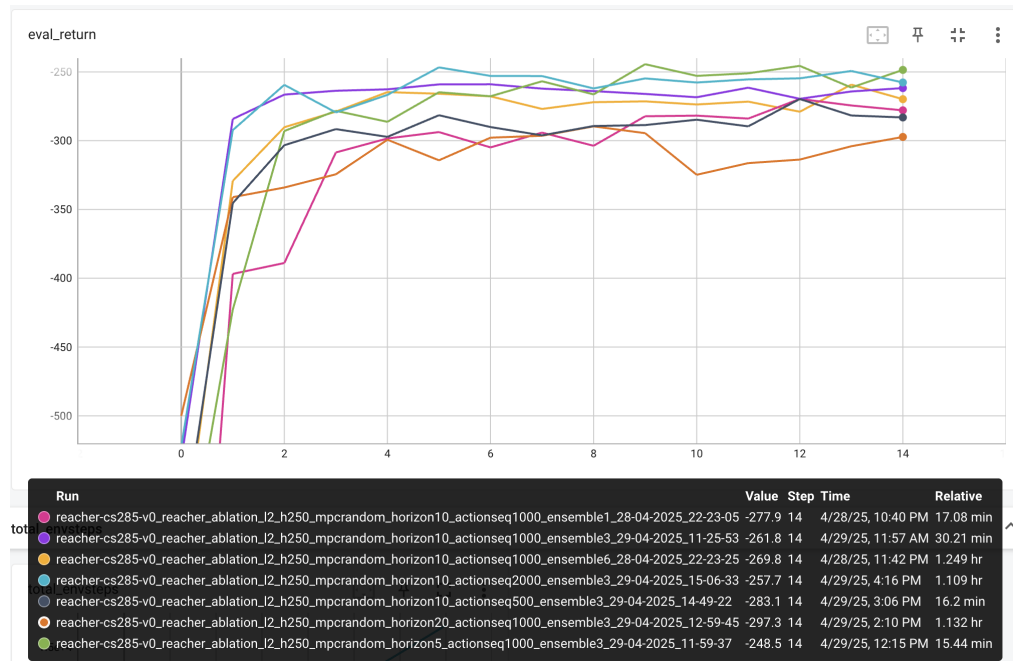


Figure 8: Eval return for Reacher.

Problem 5

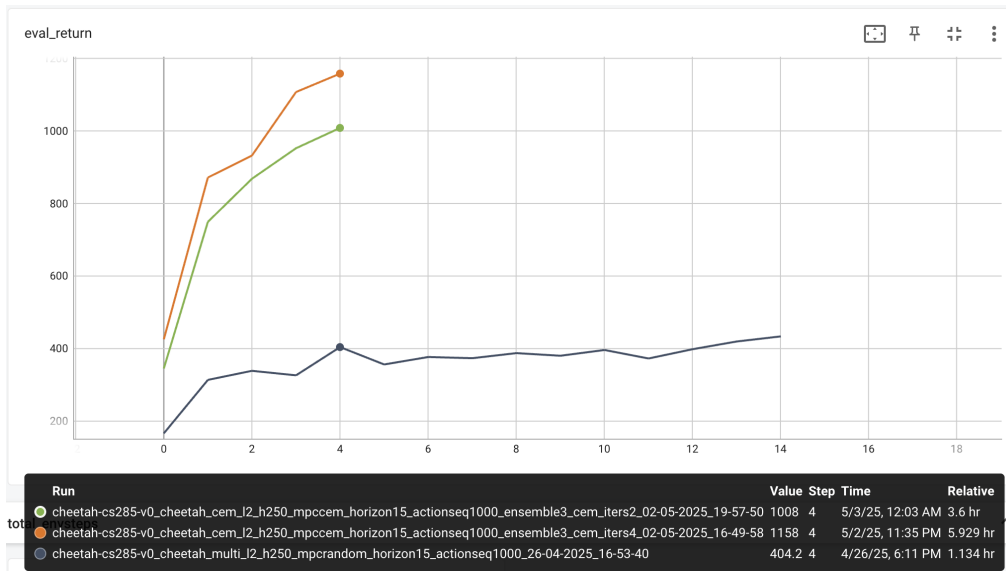


Figure 9: Eval return for Cheetah: random shooting (black), CEM for 2 iterations (green) and 4 iterations (orange). The return increases as the number of CEM iterations increases.

Problem 6

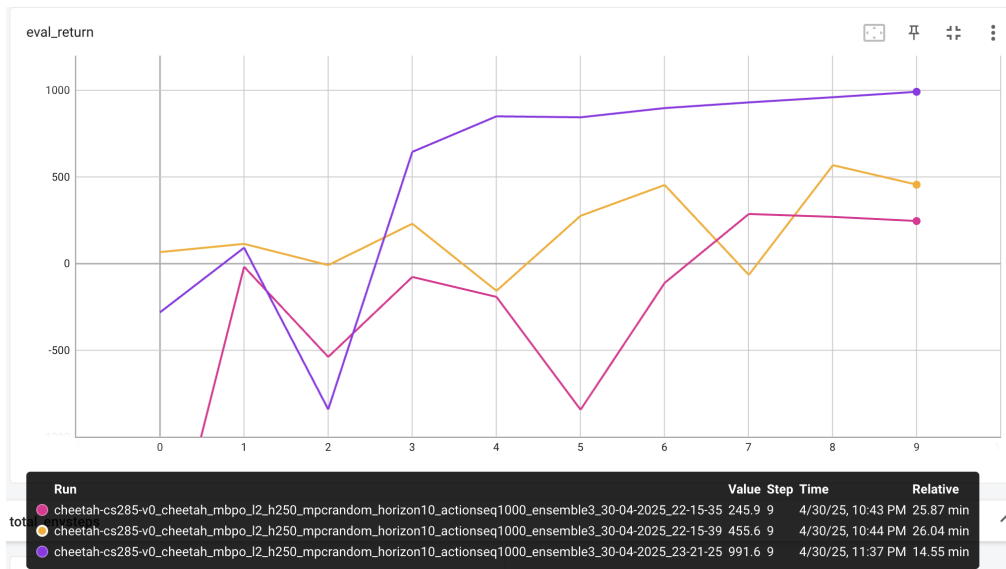


Figure 10: Eval return for Cheetah: baseline (pink), dyna (yellow), mbpo (purple)